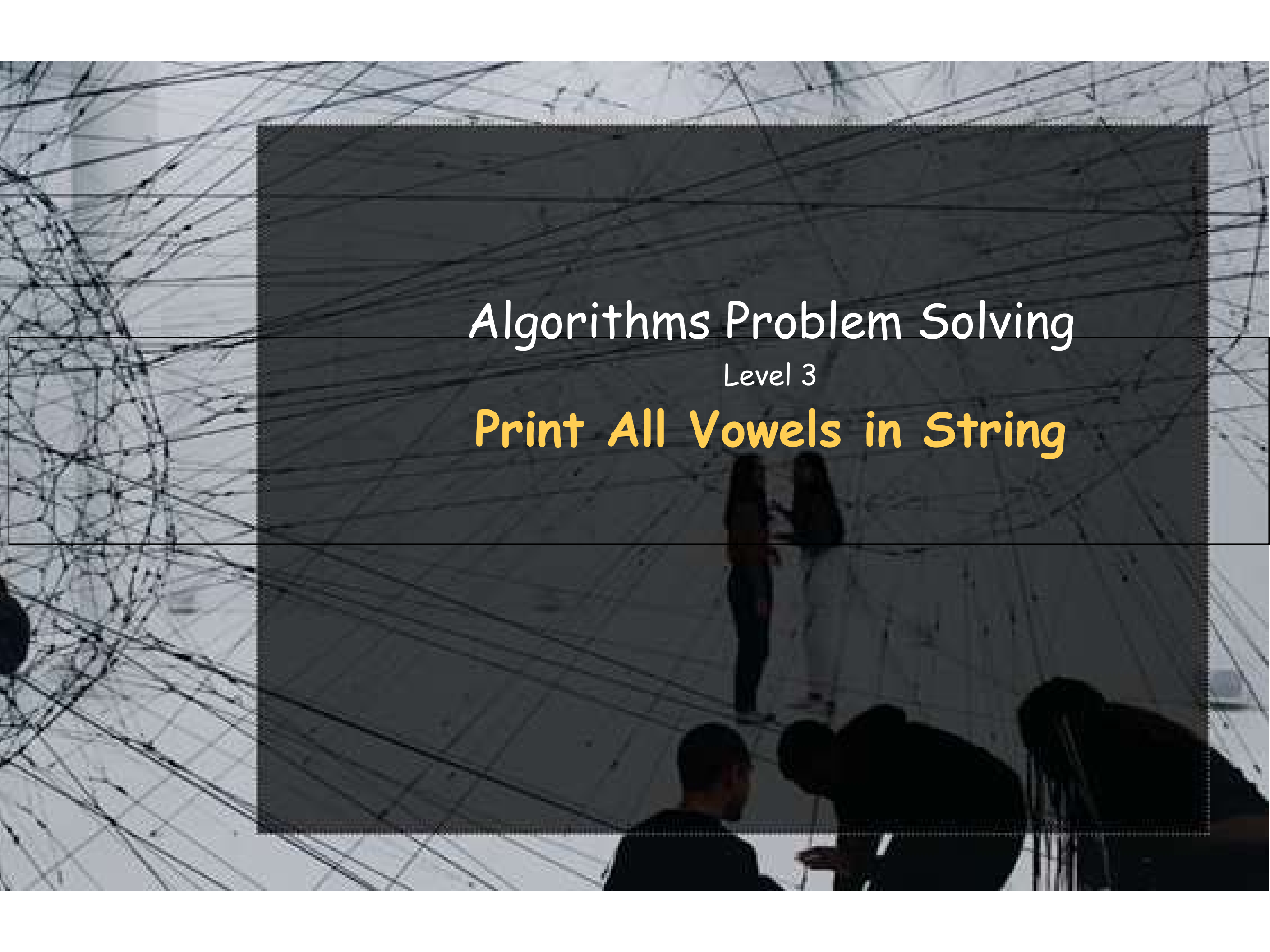


# Algorithms Problem Solving Level 3

## Print All Vowels in String



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# Definition

## Vowel Character Extraction

-  Program scans a text string and prints every vowel it encounters.
-  The vowels are displayed in the same order in which they appear in the input.

# Explanation

## Character Traversal

-  Every character in the string is inspected sequentially.
-  This guarantees that no vowel occurrence is skipped.

## Predicate Filtering

-  Each character is passed to ``IsVowel()``.
-  Only characters for which the predicate returns ``true`` are selected.

## Ordered Output

-  Matching vowels are printed immediately.
-  Their original order is therefore preserved.

# Procedures

1. Read the complete string.
2. Traverse every character.
3. Call `IsVowel()`.
4. Print each character that is identified as a vowel.

# 💡 Example

Input:

Sara Alterkawi

➡ Result

Vowels in string are:

a e i

## Note

- ◆ The function can preserve the original letter case.
- ◆ It can also normalize all printed vowels to lowercase if desired.
- ◆ This is a filtering operation rather than a counting operation.

## ⊘ Warning

- ◆ Use ``getline()`` when the input contains spaces.
- ◆ Output formatting should separate vowels clearly.
- ◆ Empty strings simply produce no vowel output.



# Comparison



## `PrintVowels()`

- ✓ Outputs the actual vowel characters.
- ✓ Preserves their occurrence order.
- ✓ Focuses on extraction and display.



## `CountVowels()`

- ✓ Returns the number of vowels.
- ✓ Does not need to output individual characters.
- ✓ Focuses on frequency calculation.

## ✦✦ Summary

- ✓ Traverses a complete input string.
- ✓ Uses ``IsVowel()'` as a filtering predicate.
- ✓ Prints every vowel occurrence.
- ✓ Preserves the original sequence of vowel appearances.

Thank you 😊